

Arezoo Rajabi

Senior AI/ML Engineer

Adjunct faculty at University of Washington

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AI/ML Researcher and Engineer with expertise in AI safety, LLM evaluation, multimodal AI, and trustworthy machine learning. Experienced building production AI systems serving 27,000+ users while designing evaluation frameworks, safety analyses, and scalable experimentation pipelines. PhD in Computer Science with publications at CCS, PETS, and AsiaCCS, bridging frontier AI research with production engineering.

EDUCATION

Ph.D. in Computer Science , <i>Oregon State University, Corvallis, Oregon, USA</i>	Sep. 2014 – June 2021
M.Sc. in Computer Engineering , <i>Sharif University of Technology, Tehran, Iran</i>	Sep. 2011 – Sep. 2013
B.Sc. in Computer Science , <i>Sharif University of Technology, Tehran, Iran</i>	Sep. 2005 – Dec. 2010

EXPERIENCE

Senior AI/ML Engineer , <i>Wells Fargo Bank, CA, USA</i>	Dec. 2022 – Present
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- *Interactive Chatbot*: Designed and deployed an enterprise LLM chatbot serving 27,000+ users, enabling question answering across technical documentation and internal knowledge bases. Developed evaluation datasets, prompt optimization strategies, and automated benchmarking pipelines to improve response quality, reduce hallucinations, and increase answer relevance through iterative offline and human-in-the-loop evaluation. Designed online quality metrics to continuously monitor model performance.
- *Video Analysis for Suspicious Behavior & Policy Violation Detection*: Led development of a multimodal AI system using Vision-Language Models for enterprise safety monitoring. Built scalable data collection and video annotation pipelines, reducing inference cost by 85% through intelligent frame selection and automated video segmentation. Designed evaluation pipelines, prompt-generation strategies, and context-aware metadata systems that improved detection precision while reducing hallucinations.
- *Applied ML Solutions*: Built predictive models for fraud detection, anomaly detection, and operational risk, performing feature engineering, model evaluation, error analysis, and threshold optimization to reduce false alarms and improve operational efficiency.

Postdoctoral Scholar , <i>NSL Lab, University of Washington, Seattle, WA, USA</i>	March 2021 – Dec. 2022
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- *Multi-Domain Trojan Detection*: Proposed and evaluated a multi-domain Trojan detection framework across text, image, and audio models, achieving 85% detection accuracy through extensive benchmarking under adversarial settings. Achieved a minimum success rate of 85% for detecting Trojan samples in text, images, and audio domains.
- *Privacy-Preserving RL Algorithm*: developed and evaluated privacy-preserving reinforcement learning algorithms, conducting extensive experiments to quantify privacy-utility tradeoffs under adversarial settings.
- *Federated Learning for Trojan Prevention*: Proposed a federated learning approach to counteract Trojan samples during the training phase.

SKILLS

Programming Languages: Python, SQL, Matlab

ML: PyTorch, Transformers, LLMs, RL Supervised/unsupervised learning techniques, MLOps

LLM & AI Safety: LLM Evaluation, Prompt Engineering, RAG, Safety Evaluation, Agentic AI, Model Fine-Tuning

Data Science: Statistical Modeling, Experimental Design, KPI Metrics Design, Visualization, Pandas, Spark, A/B Testing

PUBLICATIONS

1. **A. Rajabi**, S. Asokraj, F. Jiang, L. Niu, B. Ramasubramanian, J. Ritcey, R. Poovendran, MDTD: A Multi-Domain Trojan Detector for Deep Neural Networks, ACM Conference on Computer and Communications Security (ACM CCS), Sep. 2023.
2. J. Jia, Z. Yuan, D. Sahabandu, L. Niu, **A. Rajabi**, B. Ramasubramanian, B. Li, R. Poovendran, FLGAME: A Game-theoretic Defense against Backdoor Attacks In Federated Learning, Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS), Sep 2023 (<https://neurips.cc/virtual/2023/poster/70499>).
3. **A. Rajabi**, D. Sahabandu, L. Niu, B. Ramasubramanian, R. Poovendran, LDL: A Defense for Label-Based Membership Inference Attacks, ACM Asia Conference on Computer and Communications Security (AsiaCCS), July 2023 (**7% acceptance rate**).
4. **A. Rajabi**, B. Ramasubramanian, A. Marruf, R. Poovendran, Privacy Preserving Reinforcement Learning Beyond Expectation, Accepted in 61st IEEE Conference on Decision and Control, 2022.(<https://arxiv.org/pdf/2203.10165.pdf>).
5. **A. Rajabi**, M. Abbasi, R. B. Bobba, K. Tajik, Adversarial Images Against Super-Resolution Convolutional Neural Networks for Free, Privacy Enhancing Technology Symposium (PETS), 2022.
6. M. Abbasi, **A. Rajabi**, C. Shui, C. Gagné, R. B. Bobba, Toward Adversarial Robustness by Diversity in an Ensemble of Specialized Deep Neural Networks, Canadian Conference on Artificial Intelligence (Canadian AI), 2020. (Best Paper Award)

PATENTS

Arezo Rajabi, Dinuka Sahabandu, Luyao Niu, Bhaskar Ramasubramanian, Radha Poovendran, *LDL: A Defense for Label-Based Membership Inference Attacks*, Record of Innovation filed with CoMotion At University of Washington, Seattle Dec. 2022.

SERVICES

Adjunct Faculty at University of Washington	2022- present	Organizer at The Trojan Detection Challenge (LLM Edition), NeurIPS 2023	2023
Organizer at Trojan Detection Challenge, NeurIPS 2022			2022
Diversity co-chair at Security and Privacy Symposium			2023

AWARDS

First Place Winner at Graduate Research Showcase for Poster Presentation	2018
Cyber Resilient Energy Delivery Consortium (CREDC) Summer School Student Scholarship	2017
Student Travel Awards from Top Security Conferences (S&P, CCS, GREPSEC, and ACSAC)	